Fiscal Year:	FY 2008	**Task Last Updated:**	FY 01/28/2009
PI Name:	Archambeau, John Orin M.D.		
Project Title:	Quantification Of The Dose Response Of The Microvessel Parameters In Retina, Cortex And White Matter Following Iron-56 Irradiation And Proton Irradiation		
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Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Space Biology Special Category:	None		
PI Email:	joarcham@aol.com	**Fax:**	FY (909) 558-0825
PI Organization Type:	UNIVERSITY	**Phone:**	(909) 558-8370
Organization Name:	Loma Linda University Medical Center		
PI Address 1:	Chan Shun Pavilion		
PI Address 2:	11175 Campus St.		
PI Web Page:			
City:	Loma Linda	**State:**	CA
Zip Code:	92354	**Congressional District:**	41
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mao, Xiao (Radiation Biology Program Department of Radiation Medicine)		
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	A main concern of NASA managers is to define the risks of tissue damage of brain and eye for astronauts associated with exposure to galactic cosmic ray particle (GCR) irradiation during a proposed space mission to Mars. Research is needed to specify the tissue effects produced by the high Z, high energy (HZE) particles that are found in space and to define the dose response of these tissues. Current histopathologic methods used to evaluate tissue changes in late-responding, non-proliferative tissues such as the brain and vessel endothelium rely on observations, word descriptions and arbitrary scoring. The approach is considered to be qualitative and lacks rigor. The proposed research introduces a detailed, step-by-step stereology program into radiation biology that will quantify the dose response of tissue population changes in situ following irradiation. These methods are known to anatomists and materials scientists but have been ignored by radiation biologists until this laboratory used them to quantify the dose response of the		

Task Description:	non-proliferative retinal vasculature following single and split time-dose schedules of protons [1, 2]. The brain and retina microvessel endothelial cell population number, vessel length, and number of microvessel branching will be measured following single and split dose schedules of Iron-56 particles. The results will be compared with the changes produced by protons to define the dose response of the cortex and white matter microvessel endothelial populations and to specify the RBE quality factor of Iron-56 referenced to protons. Three-dimensional reconstructions of the microvessels of the cortex and white matter will be compared. Our objective is to utilize three dimensionsl reconstruction and stereology methods to quantify the microvessel population and vessel cell kinetic parameters in the cortex and white matter regions of the brain in an effort to explain the difference in dose response between the two regions. The significance of this research for NASA and for radiation biology is that it introduces into radiation biology a valuable quantitative technique that can be applied to quantify population morphologic, kinetic and identifiable functions in-situ following irradiation. It also defines the contribution (role) of the irradiated vasculature in producing radiation changes and the dose response of the brain and retina.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These techniques used to quantify the dose response of the retina and the brain can be applied in future translational patient studies to determine the radiation dose response.
Task Progress:	1. We used unbiased stereology to determine the population density of the pericytes and endothelial cells of the cortex and corpus callosum as well as the microvessel length density. 2. Using computer assisted 3 dimensional configurations and confocal immunohistochemical microscopy we have demonstrated the branching of cortical and corpus callosum microvessels and location of the microvessel population nuclei. 3. The microvessels of the corpus callosum are demonstrated to be nearly straight and less branching than those of the cortex. 4. Using the computer technology of our Czech and Slovak collaborators it was possible to fuse confocal small images into a composite image consisting of 9 adjacent fields of the hippocampus and adjacent corpus. 5. The dose response of the endothelial and pericyte cell density of the cortex and corpus callosum were determined following iron-56 ion and proton irradiation dose schedules. The endothelial cell population of the cortex and corpus and the microvessel length density were proliferative and increased their population. The pericyte population was not proliferative and their population decreased over a period of two years. 6. Using retinal digest techniques we determined the RBE of the iron-56 ion to be 3.45 compared to the proton density change as the standard. Conclusion. 1. We document that the dose response or cell density changes were the same using split dose schedules compared with single dose schedules of protons. This suggeststhat the dose response will not be markedly changed using fractionated irradiation 2. We show that the microvessels of the cortex were more branched, and longer than the vessels of the white matter. The cortical cell density and branchings of the vessels of the cortex were larger than the same measurements for the corpus callosum. 3. The endothelial cells of the cortex and the corpus were proliferative and increased their number over the two year observational period following proton and Iron-56 ion brain irradiation with doses of 5 to 20Gy. Whereas, pericytes showed cell loss, but (surprisingly) there was an increase in vessel length density. 4. The data confirm the concerns of NASA managers that whole brain irradiation will with 5 to 20 Gy of protons or Iron-56 irradiation will produce an increase in cell density of endothelial cells and length density of microvessel, and will produce a significant loss of pericytes.
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